

1. Administrative & Study Identification

Full Study Title: First in Human Study to Evaluate AZD9793 in Participants With Advanced or Metastatic Solid Tumours **Short Title/Acronym:** RHEA-1 **Protocol Number:** RHEA-1 **NCT Number:** NCT06795022 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD9793 (GPC3 TITAN T-cell engager / trispecific IgG1 monoclonal antibody targeting GPC3 and CD8) **Phase of Development:** Phase I/II **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety and tolerability of experimental treatment with AZD9793 in participants with advanced or metastatic solid tumors. **Secondary Objectives:** To evaluate preliminary anti-tumor activity (efficacy) in terms of Objective Response Rate (ORR), Best Objective Response (BOR), Disease Response Rate (DRR), Duration of Response (DoR), Time to Response (TTR), Progression-Free Survival (PFS), and Overall Survival (OS); to assess pharmacokinetics (PK parameters); and to evaluate immunogenicity and CD8+ T cell infiltration pre- and post-treatment.

2.2 Background & Rationale To address the unmet medical need in patients with advanced or metastatic solid tumors, specifically Hepatocellular Carcinoma (HCC), expressing Glypican-3 (GPC3). AZD9793 is a novel, first-in-class CD8-guided T cell-engager (TCE) designed to directly engage tumor-infiltrating T cells and GPC3+ tumors, offering a new targeted mechanism to attack cancer cells specifically while sparing healthy cells.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional (Open-label, non-randomised, multi-centre, first-in-human dose escalation and expansion) **Control Type:** Single-arm (Monotherapy modules) **Blinding:** Open-label **Randomization:** Non-randomised

3.2 Planned Enrollment & Duration **Target \$N\$:** 304 evaluable patients **Number of Sites:** Multi-centre (Enrolling in the US and APAC regions) **Estimated Study Start:** 2024 / 2025 (Currently Recruiting) **Estimated Primary Completion:** >2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years at the time of signing the informed consent. 2. GPC3 positive tumour as determined by a central laboratory using an analytically validated IHC assay. 3. Confirmed advanced recurrent and/or metastatic and/or unresectable HCC, which is histopathologically proven based on WHO criteria. 4. Must have at least one measurable lesion according to Response Evaluation Criteria in Solid Tumors (RECIST) v1.1. 5. Eastern Cooperative Oncology Group (ECOG) Performance status of 0-1 at screening, with a predicted life expectancy of $\geq$ 12 weeks.

4.2 Exclusion Criteria 1. Known fibrolamellar HCC, sarcomatoid HCC, or combined hepatocellular malignant cholangiocarcinoma. 2. Central nervous system (CNS) metastases or CNS pathology within 3 months prior to consent. 3. Infectious disease including active human immunodeficiency virus (HIV), and uncontrolled active systemic fungal, bacterial, or other infection. 4. History of thromboembolic event within the past 3 months prior to the scheduled first dose of study intervention. 5. Cardiac conditions as defined by the protocol.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Modular dose-escalation (Part A) and dose-expansion (Part B). Specific dose levels are determined progressively during the dose-escalation phase. **Route of Administration:** Module 1: Intravenous (IV) monotherapy. Module 2: Subcutaneous (SC) monotherapy. **Duration of Treatment:** Treatment continues until disease progression, unacceptable toxicity, or patient/investigator decision.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care algorithms for managing cancer symptoms and treatment-related toxicities. **Prohibited:** Concurrent investigational agents, live vaccines, and other systemic anti-cancer therapies outside the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, RECIST evaluation, GPC3 IHC central testing, ECOG, Safety Labs
Baseline	Day 0	First Dose (IV or SC), Safety Baseline verification
Interim	Cycle 1 and beyond	Safety Labs, DLT Assessment (up to end of Cycle 1), PK/Immunogenicity profiling, Tumor Imaging
End of Study	Follow-up	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability, quantified by the number of patients with adverse events (AEs), serious adverse events (SAEs), adverse events of special interest (AESIs), and dose-limiting toxicities (DLTs). **Measurement Method:** Clinical assessment and standard laboratory evaluations. A DLT is measured from the first dose up to the planned end of Cycle 1 (the DLT assessment period).

7.2 Secondary & Exploratory Endpoints * Preliminary Efficacy: Objective Response Rate (ORR), Best Objective Response (BOR), Duration of Response (DoR), Progression-Free Survival (PFS), and Overall Survival (OS) as assessed by RECIST v1.1. * Pharmacokinetics (PK parameters) and Immunogenicity. * CD8+ T cell infiltration pre- and post-treatment.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous clinical collection of AEs and SAEs from the time of informed consent through the defined follow-up period. **Serious Adverse Events (SAEs):** Protocol-defined timeline, typically reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** A safety review committee (SRC) will evaluate DLTs during the dose-escalation phase to determine the transition to the next dose cohort and recommended phase 2 dose.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all AE/SAE/DLT evaluations. Efficacy analysis set for ORR, PFS, and OS. **Sample Size Justification:** $N=304$. No formal statistical hypothesis is proposed; all variables will be reported descriptively to guide further development phases. **Handling of Missing Data:** Standard algorithms for early-phase oncology studies (e.g., treating participants with missing efficacy scans as non-responders for ORR).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of safety data, DLTs, and protocol adherence. **Audit Trail:** Robust electronic Case Report Form (eCRF) locking, independent verification of RECIST scans, and central lab validation of GPC3 IHC assay results.

11. Study Identifiers & Governance

Primary ID: RHEA-1 **Registry Number:** NCT06795022 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** RHEA-1: First-in-human (FIH) study of AZD9793 **Official Regulatory Title:** First in Human Study to Evaluate AZD9793 in Participants With Advanced or Metastatic Solid Tumours

12. Clinical Framework (Oncology / HCC Specific)

Primary Condition: Advanced or Metastatic Solid Tumors / Hepatocellular Carcinoma (HCC) **Diagnosis Threshold:** GPC3 positive tumour expression **Medical Methodology:** T cell-engaging immunotherapy (TCE) **Optimization Target:** CD8-guided T cell-engagement to elicit targeted anti-tumor activity

13. Study Procedures & Methodology

Management Pathway: Stepwise dose escalation (Part A) followed by dose expansion (Part B). **Education Protocol (HCP):** Site initiation visits and detailed training on recognizing and managing immune-mediated toxicities/DLTs, as well as distinct administration protocols (IV vs. SC). **Education Protocol (Patient):** Informed consent detailing first-in-human risks and strict follow-up schedules. **Quality Audit Mechanism:** Centralized laboratory validation for GPC3 positivity and strict adherence tracking for RECIST imaging schedules.

14. Observation & Follow-Up Schedule

Total Duration: Continued until disease progression, unacceptable toxicity, or study withdrawal. **Onsite Visit Cadence:** Intensive monitoring during Cycle 1 (DLT window), transitioning to regular day 1 visits of subsequent cycles. **Remote Monitoring Frequency:** Routine data/safety monitoring checks per institutional Phase I standards. **Checklist Review Interval:** Regular tumor imaging assessments (e.g., every 8-12 weeks) to review RECIST response (PFS and DoR).

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator			[Insert Name]			____		[DD-MMM-YYYY]								
	Clinical Operations Lead			[Insert Name]			____		[DD-MMM-YYYY]								
	Lead Statistician			[Insert Name]			____		[DD-MMM-YYYY]								